

CS474 – HW1

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1. (10 points) Consider the perceptron in two dimensions: $h(x) = \text{sign}(w^T x)$ where $w = [w_0, w_1, w_2]^T$. Technically, x has three coordinates, but we call this perceptron two-dimensional because the first coordinate is fixed at 1.

a) Show that the regions on the plane where $h(x) = +1$ and $h(x) = -1$ are separated by a line. If we express this line by the equation $x_2 = ax_1 + b$, what are the slope a and intercept b in terms of w_0, w_1, w_2 ?

b) Draw pictures for the cases $w = [1, 2, 3]^T$ and $w = -[1, 2, 3]^T$. Label the positive and negative prediction regions on the pictures.

1A)

$w^T x = 0$ is the decision boundary where the sign flips from positive to negative.

Therefore, $w_0 + w_1 x_1 + w_2 x_2 = 0$

Then solve for x_2 :

$$w_2 x_2 = -w_0 - w_1 x_1$$

$$w_2 x_2 / w_2 = (-w_0 - w_1 x_1) / w_2$$

$$x_2 = -(w_0 / w_2) - (w_1 / w_2) x_1$$

Final equation is $x_2 = -(w_1 / w_2) x_1 - (w_0 / w_2)$

So, $a = -w_1 / w_2$ and $b = -w_0 / w_2$

1B)

Case A: $w = [1, 2, 3]^T$

$$w_0 + w_1 x_1 + w_2 x_2 = 0$$

$$1 + 2x_1 + 3x_2 = 0$$

Then solve for x_2 :

$$3x_2 = -2x_1 - 1$$

$$x_2 = -x_1 -$$

$$3x_2 = -2(0) - 1 \rightarrow x_2 = -1/3$$

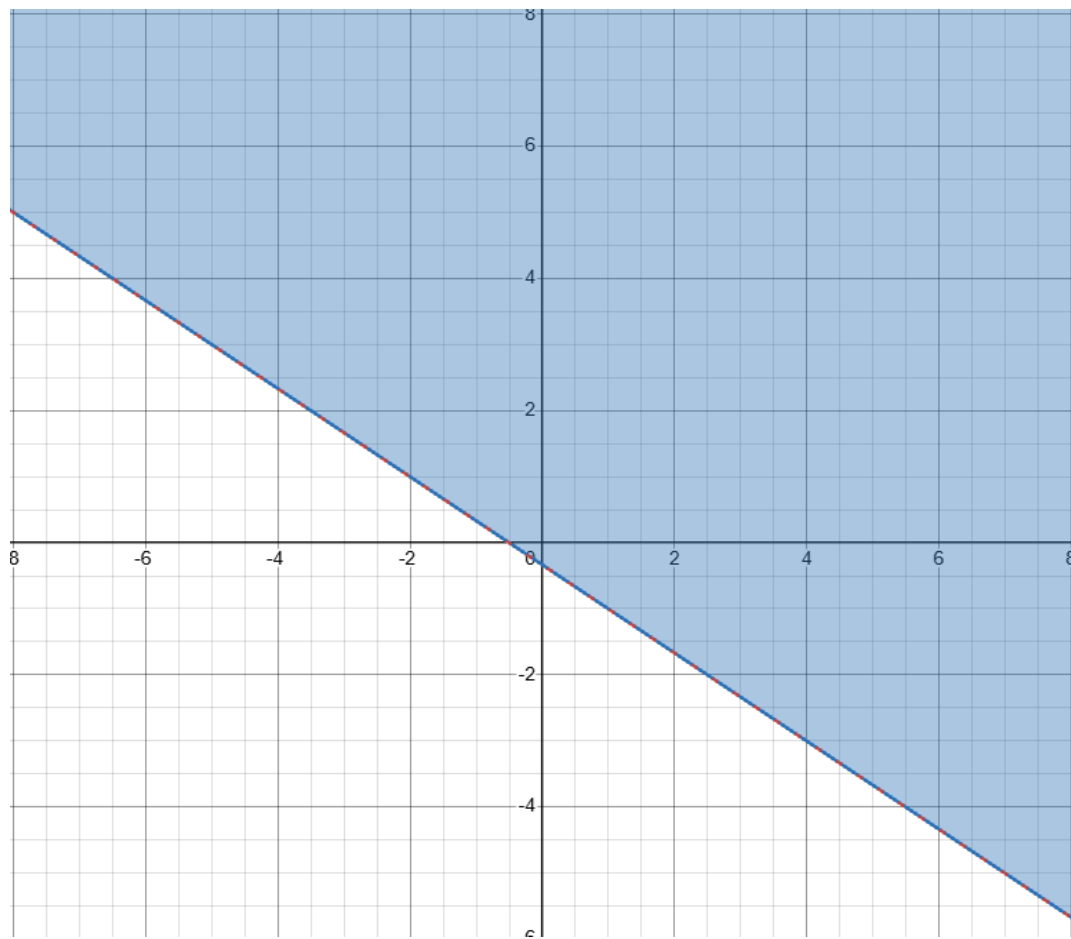
When $x_1 = 0$, $x_2 = -$

$$3(0) = -2x_1 - 1 \rightarrow 1 = -2x_1 \rightarrow x_1 = -1/2$$

When $x_2 = 0$, $x_1 = -1/2$

Testing $(0, 0)$ on line makes it $1 + 2(0) + 3(0) = 1$, so $(0, 0)$ side is $+1$ (positive)

Two points for the decision boundary line are $(0, -)$ and $(-1/2, 0)$



Case B: $w = [-1, -2, -3]^T$

$$w_0 + w_1x_1 + w_2x_2 = 0$$

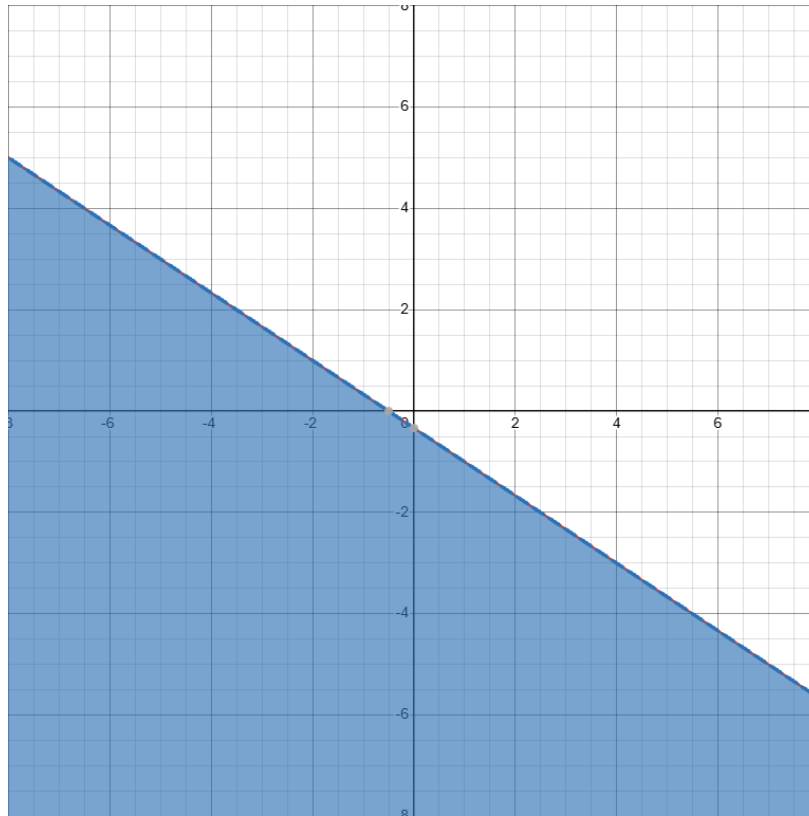
$$-1 - 2x_1 - 3x_2 = 0$$

Solve for x_2 :

$$-3x_2 = 2x_1 + 1$$

$$x_2 = -2/3x_1 - 1/3$$

Testing $(0, 0)$, $-1 - 2(0) - 3(0) = -1$, so $(0, 0)$ side is negative



2A:

Per-example loss

$$n(w) = \ln(1 + e^{-y_n w^T x_n})$$

$$E(w) = \frac{1}{N} \sum_{n=1}^N n(w)$$

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Chain rule for $n(w) = \ln(g(w))$ where $g(w) = 1 + e^{-y_n w^T x_n}$

Then $n(w) = \frac{1}{g(w)} * g'(w)$

Differentiate $g(w) = 1 + e^{-y_n w^T x_n}$ so that the 1 turns into a 0

$$g'(w) = (-e^{-y_n w^T x_n})$$

Then chain rule it again as derivative of e^u is $e^u u$

And $u(w) = -y_n w^T x_n$, so

$$(e^{-y_n w^T x_n})' = e^{-y_n w^T x_n} * (-y_n x_n)$$

Then differentiate $-y_n w^T x_n$ with respect to w

$$w^T x_n = x_n^T w$$

So, $(-y_n w^T x_n)' = -y_n x_n$

$$g'(w) = e^{-y_n w^T x_n} (-y_n x_n)$$

$$n'(w) = \frac{1}{(1 + e^{-y_n w^T x_n})} * e^{-y_n w^T x_n} (-y_n x_n)$$

With $z = ynw^T x_n$, simplify above so that it is just
 $e^{-z}/(1 + e^{-z}) = 1/(1 + e^z)$

Plugging it back in gets:

$$n(w) = -(y_n x_n)/(1 + e^{(ynw^T x_n)})$$

Final Derivative Answer:

$$E(w) = -1/N \sum_{n=1, N} y_n x_n / (1 + e^{(ynw^T x_n)})$$

2B:

Even though the logistic regression uses a nonlinear sigmoid, the place where the prediction flips is still a straight line. Logistic regression is a linear classifier despite using a nonlinear sigmoid. The prediction rule states that the prediction is 1 if $(w^T x) \geq 0.5$ else predict -1. And $(z) = 1/(1+e^{-z})$.

Taking $(w^T x) = 0.5$, we can use sigmoid property of it being strictly increasing and $(0) = 1/(1 + e^0) = 0.5$. Therefore, $(w^T x) = 0.5$ and $w^T x = 0$.

$w^T x = 0$ is linear function of x and the logistic regression function depends on it for a prediction. Therefore the decision boundary is a linear function.

2C:

Decision boundary exists where $(w^T x) = 0.9$

So need to do the same thing as 2B where $(z) = 1/(1+e^{-z}) = 0.9$

$$1/(1+e^{-z}) = 0.9$$

$$1 + e^{-z} = 1/0.9 \text{ or } 10/9$$

$$e^{-z} = 1/9$$

$$-z = \ln(1/9) \quad (\ln 1/9 = -\ln 9)$$

$$-z = -\ln 9 \rightarrow z = \ln 9$$

Therefore the boundary occurs at $w^T x = \ln 9$. This shows that the logistic regression function is still linear despite using a sigmoid function, as similarly to the previous problem, the decision boundary is a linear line.

2D:

The essential property is that the sigmoid is invertible and monotonic, so probability maps to linear score regarding the linear decision boundary. The sigmoid is always increasing as the input increases, comparing the output probability to a decision to a point c is the same as comparing the input $w^T x$ to a point. So the boundary is still where $w^T x$ equals a constant line.

3:

Let e_i be a standard basis vector, which has a singular 1 in position i and 0 for every other value in the vector. $e_i = [1, 0, 0, 0, 0]$

Now compute Y by stacking the rows $y_i^T \rightarrow Y = (i=1, n) e_i y_i^T$

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Now $e_i y_i^T$ is matrix that has all zeros except row i , which is y_i^T

Now multiply both sides by X to get $XY = X (i=1, n) e_i y_i^T$

Then insert X into sum $\rightarrow XY = (i=1, n) X (e_i y_i^T)$

Next compute $X e_i$, which will only take the i th column of X since e_i is a standard basis vector. So

it will be $X_{ei} = x_i$.

Then, therefore $XY = (i=1, n) x_i y_i^T$

4:

$C = 2$ so there are two different classes. Softmax reduces to sigmoid as shown below

$$\begin{aligned}\text{sigmoid}(y) &= 1/(1 + e^{-y}) \\ \text{softmax}(y_j) &= e^{y_j} / (e^{y_1} + e^{y_2})\end{aligned}$$

$$C = 2 \rightarrow \text{softmax}(y_1) = e^{y_1} / (e^{y_1} + e^{y_2}) \text{ and } \text{softmax}(y_2) = e^{y_2} / (e^{y_1} + e^{y_2})$$

$$\begin{aligned}\text{softmax}(y_1) &= e^{y_1} / (e^{y_1} + e^{y_2}) \\ e^{y_1} / (e^{y_1} + e^{y_2}) \\ \text{Divide by } e^{y_1} \text{ for numerator and denominator} \\ (e^{y_1}/e^{y_1}) / (e^{y_1} + e^{y_2})/e^{y_1} &= 1/(1 + e^{(y_2 - y_1)})\end{aligned}$$

$$\begin{aligned}e^{(y_2 - y_1)} &= e^{-(y_1 - y_2)} \\ \text{So now it is } 1/(1 + e^{-(y_1 - y_2)}) \\ \text{This format matches the sigmoid function needed.} \\ (t) &= 1/(1 + e^{-t})\end{aligned}$$

$$\text{softmax}(y_2) = e^{y_2} / (e^{y_1} + e^{y_2}) = 1/(1 + e^{y_1 - y_2}) = e^{-(y_1 - y_2)} = 1 - e^{-(y_1 - y_2)}$$

$$\text{softmax}(y_1) = e^{y_1} / (e^{y_1} + e^{y_2}) = 1 - e^{-(y_1 - y_2)}$$